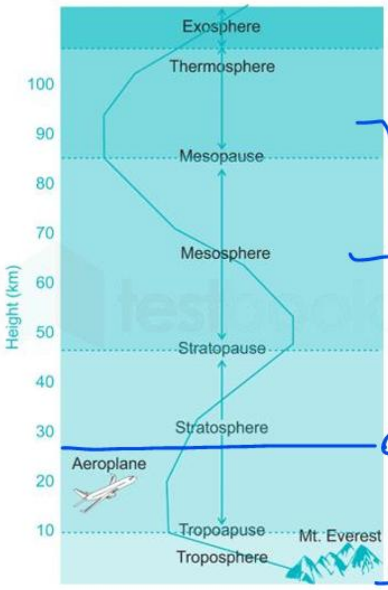
Lecture 5 and 6

Disclaimer
These notes for Lectures 5 and 6 are *not comprehensive*. They include only a selected compilation of certain important segments such as winds, maps, and a few other key points. The detailed conceptual discussions, examples, and explanations covered during the classes are to be **updated and rewritten by the aspirants in their own words** for better understanding and long-term retention.

अस्वीकरण (Disclaimer)
यह लेक्चर 5 एवं 6 से संबंधित नोट्स *संपूर्ण (comprehensive) सामग्री नहीं है।* यहाँ केवल कुछ महत्वपूर्ण भागों – जैसे हवाएँ, मानचित्र, तथा अन्य चयनित बिंदुओं – का संक्षिप्त संकलन प्रस्तुत किया गया है। कक्षा में हुई विस्तृत वैचारिक चर्चा, उदाहरण, व्याख्या तथा अवधारणात्मक भागों को अभ्यर्थियों द्वारा **अपने शब्दों में** पुनर्लिखित एवं अपडेट किया जाना अपेक्षित है, ताकि विषय की बेहतर समझ विकसित की जा सके।

$\boxed{\text{Trust Me Thermal Ex}}$



Ionosphere :- संचरात्मक x (structural) → functional कार्यात्मक

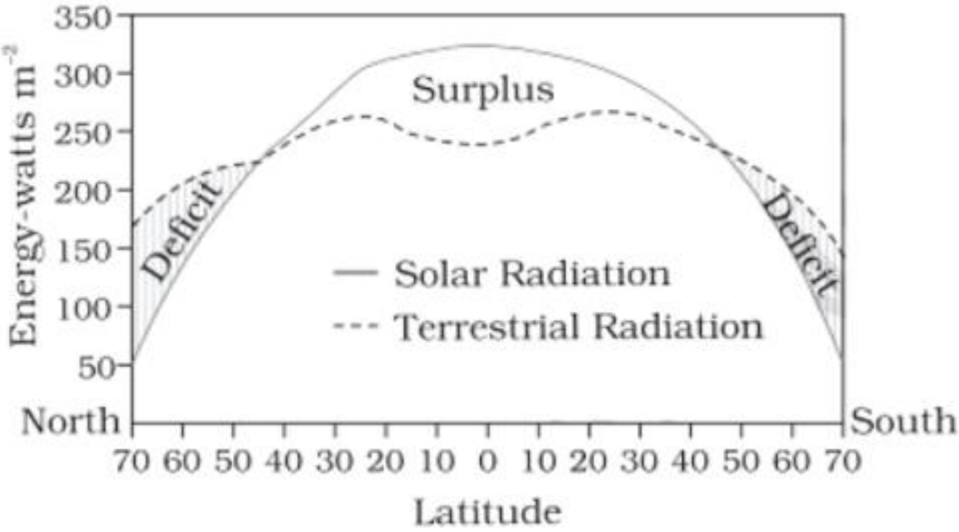
Ozone (15-20km)

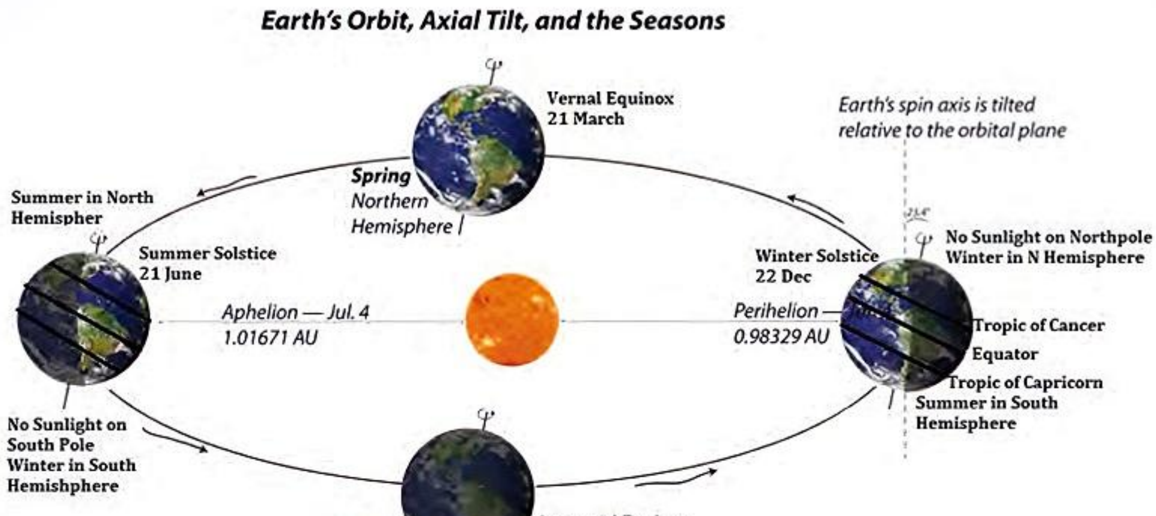
↑ -60°deg.

Lapse rate → प्रति 1 km → कितना तापमान गिरा ? -6.5

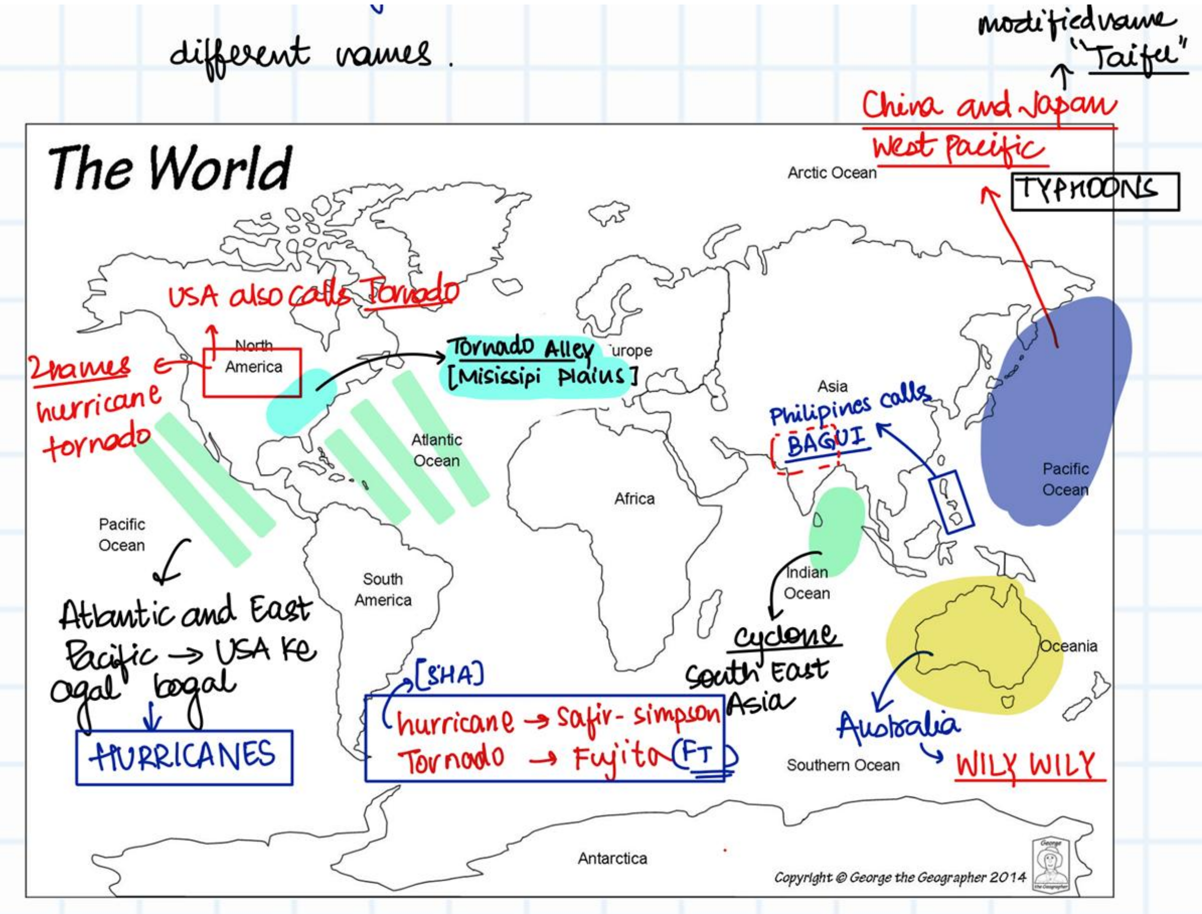
Layer	Altitude Range	Temperature Nature	Key Phenomena / Exam Relevant Facts
Troposphere (lowest layer – भूतल के सबसे पास)	8 to 18 km; take 13 km average	Temperature decreases with height (ऊँचाई बढ़ने पर तापमान कम होता है). Normal Lapse Rate (LR) = ~6.5°C per km (लैप्स रेट - ऊँचाई के साथ गिरावट दर)	Almost all weather [मौसम], clouds [बादल], rainfall [वर्षा], storms [तूफ़ान] all occur here.
Stratosphere	~12 km – 50 km	Temperature increases (ऊपर जाते-जाते तापमान बढ़ने लगता है) because of Ozone absorption (ओज़ोन द्वारा सौर-UV का अवशोषण). Goes near Zero at top.	Ozone layer exists at ~15–20 km (ओज़ोन परत). Planes fly here (No turbulence – कोई अशांति नहीं - smooth flight दृश्यता स्पष्ट रहती है).
Mesosphere	~50 km – 85 km	Temperature again decreases (ये सबसे cold layer है) ~ –90°C. Mesopause is coldest boundary region (मेज़ोपॉज सबसे सर्द सीमा क्षेत्र).	Meteors burn here (shooting stars) [उल्कापिंड यहीं जलते हैं]
Thermosphere	~85 km – 600 km	Temperature increases drastically up to 1500°C+ (बहुत तेजी से गर्म)	Ionosphere lies mainly here (आयनोस्फ़ीयर - charged particles का क्षेत्र). Auroras occur here [ध्रुवीय प्रकाश]. ISS orbits in lower Thermosphere.
Exosphere (outermost layer)	~600 km – 10,000 km	Gradual temperature variation (धीरे-धीरे variation)	Merges into outer space (अंतरिक्ष में विलीन). Most satellites operate here (उपग्रह संचालन क्षेत्र).

Ionosphere: Struc X Functional ✓ reflects Infrared waves

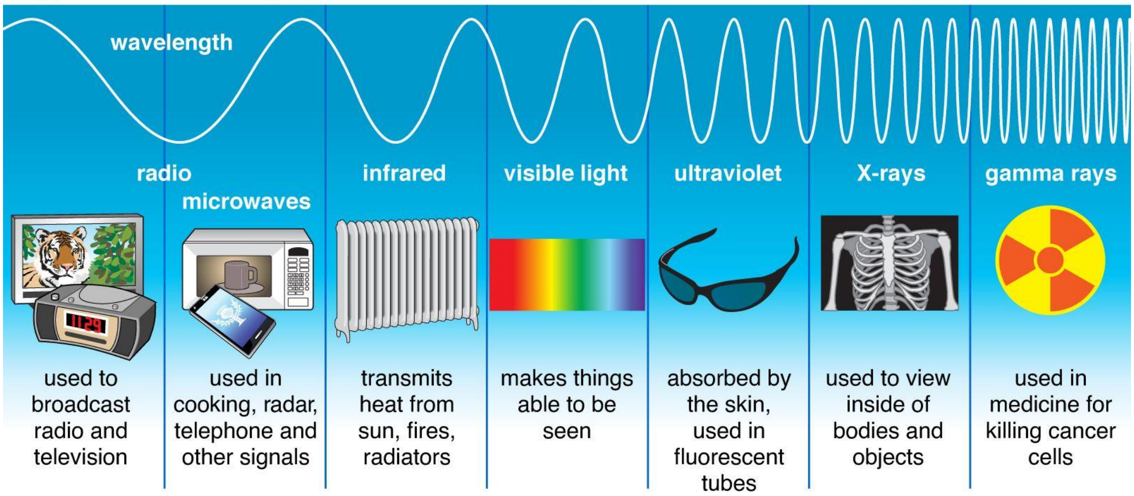

Energy-watts m⁻²
350
300
250
200
150
100
50
Surplus
Deficit
Deficit
Solar Radiation
Terrestrial Radiation
North
South
70 60 50 40 30 20 10 0 10 20 30 40 50 60 70
Latitude


Earth's Orbit, Axial Tilt, and the Seasons
Vernal Equinox
21 March
Spring
Northern
Hemisphere
Earth's spin axis is tilted
relative to the orbital plane
Summer in North
Hemispher
Summer Solstice
21 June
Winter Solstice
22 Dec
No Sunlight on Northpole
Winter in N Hemisphere
Aphelion — Jul. 4
1.01671 AU
Perihelion —
0.98329 AU
Tropic of Cancer
Equator
Tropic of Capricorn
Summer in South
Hemisphere
No Sunlight on
South Pole
Winter in South
Hemishphere
Fall
Northern
Hemisphere
Autumnal Equinox
23 Sept

different names.

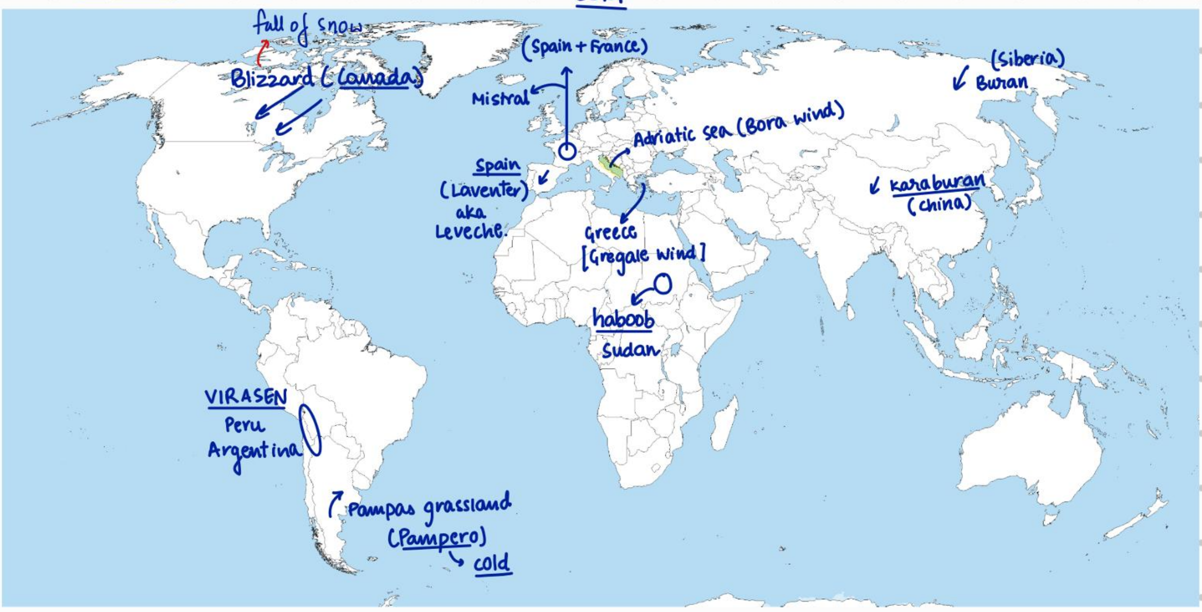


Types of Electromagnetic Radiation

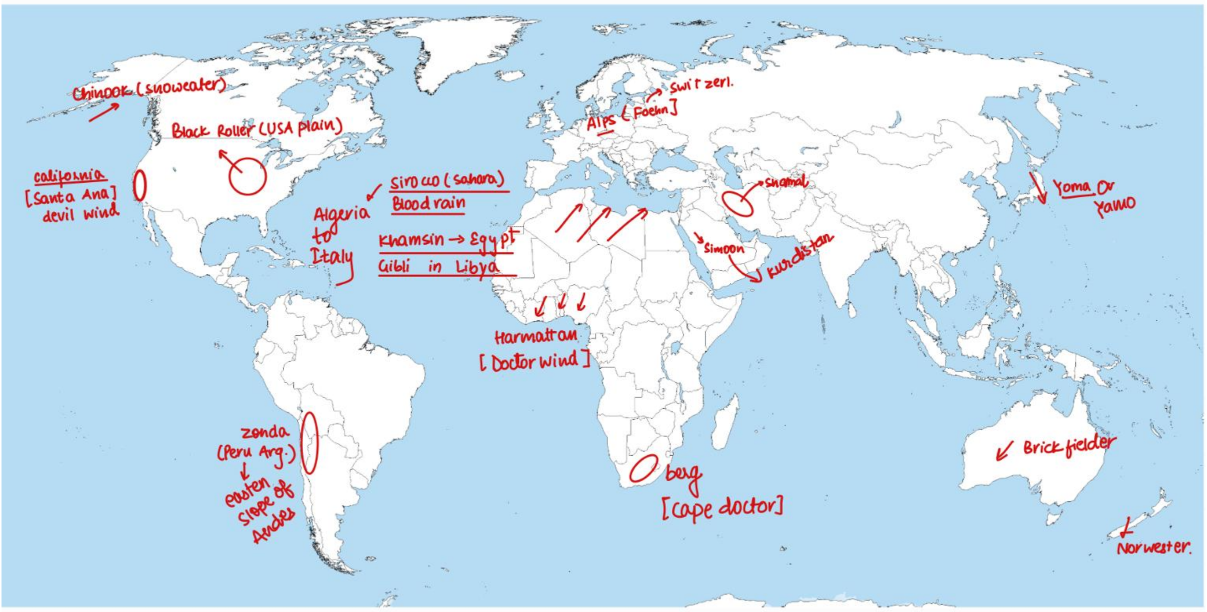


Winds

cold



Warm.



India's norwester
= KalBaishakhi

TROPICAL CYCLONE (उष्णकटिबंधीय चक्रवात)

A Tropical Cyclone is an intense **warm-core (गर्म-कोर)** low-pressure system that forms over warm tropical oceans. It derives its energy from the **latent heat of condensation (संघनन की गुप्त ऊष्मा)** released by rising moist air. These systems generally develop between **5°–20° latitude (अक्षांश)** where the **Coriolis force (कोरिओलिस बल)** is strong enough to initiate rotation.

Conditions Required

- **Sea Surface Temperature (समुद्री सतह तापमान)** above **26–27°C**
- High **humidity (आर्द्रता)** and strong **convection (संवहन)**
- Low **vertical wind shear (ऊर्ध्वाधर पवन-कतराव)**
- Pre-existing **low-pressure (निम्न दाब)** area
- Coriolis force away from equator

Structure

- **Eye (आँख)** – calm centre
- **Eyewall (आईवाल)** – strongest winds and heaviest rain
- **Rainbands (वर्षा-पट्टी)** – spiralling cloud bands

The cyclone remains symmetric and circular with a distinct **warm core (गर्म-कोर)**.

Movement & Life Cycle

- Generally moves **west–northwest (पश्चिम–उत्तरपश्चिम)** under trade winds.
- Weakens rapidly after **landfall (भूमि-स्पर्श)** because it loses moisture supply.
- Life span: 3–7 days.

Impacts

- Very high **wind speed (पवन वेग)**
- **Storm surge (उफान तरंग)** causing coastal flooding
- **Torrential rainfall (मूसलाधार वर्षा)**
- Destruction of infrastructure, agriculture, and habitation

Naming & Regions

- Indian Ocean cyclones are named by IMD.

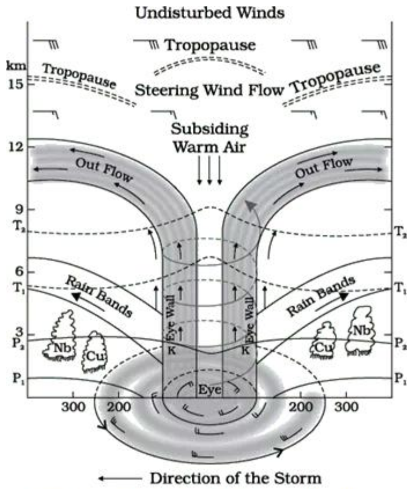


Fig: Vertical section of the tropical cyclone

TEMPERATE CYCLONE (समशीतोष्ण चक्रवात)

A Temperate Cyclone, also called an **Extratropical Cyclone (बाह्य-उष्णकटिबंधीय चक्रवात)** or **Frontal Cyclone (मोर्चीय चक्रवात)**, is a large **cold-core (शीत-कोर)** low-pressure system that forms in the **mid-latitudes (मध्य अक्षांश)** between **30°–60°**. It develops due to the interaction of contrasting **air masses (वायु-दल)** — typically cold polar air and warm tropical air — creating distinct **fronts (मोर्चे)**.

Conditions Required

- Strong **temperature gradient (तापीय अंतर)** between warm & cold air
- Presence of **fronts (मोर्चे)** — warm front, cold front, occluded front
- Influence of **Jet Streams (जेट स्ट्रीम)**
- Mid-latitude **westerlies (पश्चिमी पवन)** guiding movement

Structure

- **Cold-core (शीत-कोर)** low pressure
- Comma-shaped cloud pattern
- Well-developed **warm front (उष्ण मोर्चा)**, **cold front (शीत मोर्चा)**, **occluded front (अवरोधित मोर्चा)**
- Wide area of **frontal uplift (मोर्चीय उत्थापन)** producing widespread weather

Movement & Life Cycle

- Moves **west to east (पश्चिम से पूर्व)** under westerlies
- Strongest during **winter & spring (शीत व वसंत ऋतु)**
- Life span: several days to a week
- Weakens when temperature contrast reduces

Weather Associated

- Widespread **rainfall (वर्षा)** and **snowfall (हिमपात)**
- **Blizzards (शीत तूफ़ान)** in colder regions
- Cloudy, unsettled weather along fronts
- Moderate to strong surface **winds (पवन)**